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**Masters Program in CS : AI and ML
(AlmaBetter)**

**Product Dissection and ERD of
PhonePE App**

Github Link:-

Problem Statement

Product Dissection for top leading Platforms

Welcome to this case study on dissecting and designing products for top leading platforms. In this case study, you will delve into the intriguing world of schema design for a prominent platform of your choice. Your task is to choose a top leading platform, research its features, and meticulously craft a schema design that encapsulates the essence of its functionality. By focusing on key entities, attributes, and relationships, you will gain invaluable insights into how data architecture drives the platform's effectiveness.

Step 1: Choose a Leading Platform

Select a leading platform of your choice, which could span various domains such as social media, e-commerce, finance, or any other industry. This choice will form the foundation of your exploration into its schema design.

Step 2: Research:

Thoroughly research the platform you have selected. Investigate its core features, functionalities, and user interactions. Identify the top features that define its user experience and contribute significantly to its popularity.

Step 3: Product Dissection and Real World Problems solved by the platform

In this step, you will meticulously analyse the platform's standout features and how they provide innovative solutions to real-world challenges. By identifying key functionalities that resonate with users, you'll unravel how the platform effectively addresses problems and enhances user experiences. This dissection will serve as the foundation for understanding how the schema design aligns with the platform's core objectives.

Step 4: Case Study on the real world problems and approach to solving them

In this pivotal step, you will expand on the real-world challenges uncovered in Step 3 through a comprehensive case study. Delve into specific instances where users encountered difficulties and showcase how the platform's unique features provided effective solutions. By dissecting the approach taken by the platform to overcome these challenges, you'll gain a deeper appreciation for the platform's user-centric design philosophy and how it shapes the schema design.

Step 5: Schema Design Based on Top Features

Based on the features you have identified, craft a schema design that reflects the platform's data structure. Focus on the key entities, attributes, and relationships that underpin the chosen features. Your schema should capture the essence of how the platform organises and utilises its data.

Step 6: Rationale Behind the Design

While creating the schema design, consider the rationale behind the platform's choices. Reflect on why certain entities and relationships were chosen and how they align with the platform's goals. This will help you understand the strategic decisions driving the schema's architecture.

Step 7: Create an ER Diagram

Utilise tools like the Miro platform or similar applications to create an illustrative Entity-Relationship (ER) diagram. This diagram should vividly depict the entities, attributes, and relationships present within your schema design. The ER diagram will serve as a visual representation of your insights.

Step 8: Presentation of Findings

Present your findings in a clear and concise manner. Showcase your understanding of how the schema design impacts the platform's functionality and user experience. Explain how your chosen features are integrated into the schema and how the schema's structure supports the platform's objectives.

Task Details:

1. **Answer Submission:** Your submission should include well-structured solutions for all provided questions related to product schema designs.
2. **Video Creation:** Create an informative and engaging video where you thoroughly explain the Case Study.
3. **Depth and Clarity:** Ensure your solutions are detailed and showcase your understanding of product schema design principles. Similarly, in the video, provide clear explanations that are easy to understand for a wide audience.
4. **Creativity Encouraged:** You are welcome to utilise visuals, diagrams, or creative elements to enhance the clarity and impact of your explanations.

Note:

1. Duplicate this document and proceed to write your solutions and prepare your video.
2. Include the video link in this document before final submission.

Best of luck in completing this project and showcasing your prowess in dissecting and designing product schema for leading platforms! **For reference, we have also conducted a case study on Instagram, which you can find below. This case study will provide you with valuable insights into how schema design plays a pivotal role in shaping the functionality and success of a prominent platform.**



Product Dissection for PhonePe

Company Overview:

PhonePe, established in 2016 and proudly homegrown in India, is a revolutionary online payments platform that has revolutionized the way Indians pay. PhonePe Pvt. Ltd., an owner entity that was once a subsidiary of Flipkart and subsequently spun off from Walmart's Indian subsidiary, PhonePe is more than a fintech app; it is the face of India's digital revolution.

Consistent with the vision of the **Unified Payments Interface (UPI)**, a revolutionary initiative launched by the **National Payments Corporation of India (NPCI)**, PhonePe facilitates over half a billion Indians to conduct transactions effortlessly, securely, and in real time. With a gamut of services ranging from money transfers, bill payments, investments, and insurance, PhonePe has become an indispensable part of the lifestyle of the contemporary Indian citizen. PhonePe is a **home-grown symbol of India's standalone fintech revolution**.





Product Dissection and Real-World Problems Solved by PhonePe

PhonePe, India's number one digital payment app, was created not just to go digital with payments but also to solve everyday financial problems of crores of people. From sending money to a relative in a matter of seconds to paying an electricity bill from a far-off village, PhonePe's ease and robust infrastructure are making it a powerful force in India's financial inclusion revolution.

Problem 1: Unavailability of Instant Banking

Real-World Challenge:

Millions of Indians, particularly in rural India, had limited access to banking services. Going to banks or ATMs was time-consuming, and remitting money involved delays and paperwork.

PhonePe Solution:

Basically, PhonePe is leveraging the Unified Payments Interface (UPI) to allow users to send and receive money in real time by using a phone number or UPI ID. Users can connect their bank accounts with minimal clicks, thus eliminating Indian Financial System Codes (IFSC) or account numbers. With peer-to-peer and bank-to-bank transactions simplified, PhonePe essentially brings banking to even the most remote users.

Problem 2: Cumbersome Utility Bill Payments

Real-World Challenge:

E-bill payments such as electricity, water, gas, and DTH once needed physical trips to the service providers or waiting in multi-hour lines.

PhonePe Solution:

PhonePe consolidates thousands of billers onto its platform, thereby enabling users to pay any bill directly from anywhere in India. With auto-bill fetching, reminders, and secure payment gateways, the platform ensures users are never missed due dates. This effectively addresses a big issue for both rural and urban users, thereby transforming the bill payment process in India.

Problem 3: Unsafe Cash Transactions for Small Merchants

Real-World Challenge:

India's large cash economy is a major security threat and source of inefficiency, especially for small traders and street vendors who struggle to receive digital payments.

PhonePe Solution:

With its QR code-based payment system for merchants, even the smallest vendor can now accept UPI payments without a swiping machine or fees. PhonePe for Business app provides merchants with real-time transaction notifications, a daily report, and zero setup charges. This step is in line with India's vision of creating a cashless economy and deepening financial inclusion for small and medium enterprises.

Problem 4: Insufficient Awareness and Accessibility of Investment Opportunities

Real-World Challenge:

For years, investing in gold, mutual funds, or insurance appeared to be complicated and elitist. There was no easy way for the common fellow to participate.

PhonePe Solution:

With a minimal interface, PhonePe enables users to purchase digital gold, invest in mutual funds, and buy insurance, all within the app. No forms. No technical terminology. By democratizing access to wealth-creating instruments, PhonePe empowers the average Indian to plan their financial future, making investing as straightforward as recharging a mobile number.

Problem 5: Encouraging Day-to-Day Transactions

Real-World Challenge:

Most financial transactions are tedious and not very interesting, leading to low user engagement.

PhonePe's Solution:

Through the use of scratch cards, rewards, and cashbacks to consumers, PhonePe increases the feeling of reward for every transaction. Such game-like elements encourage repeat usage, build customer loyalty, and add a dash of amusement to simple money transfers and bill payments.

Problem 6: Over-Dependence on Cashbacks to Foster Retention

Real-World Challenge:

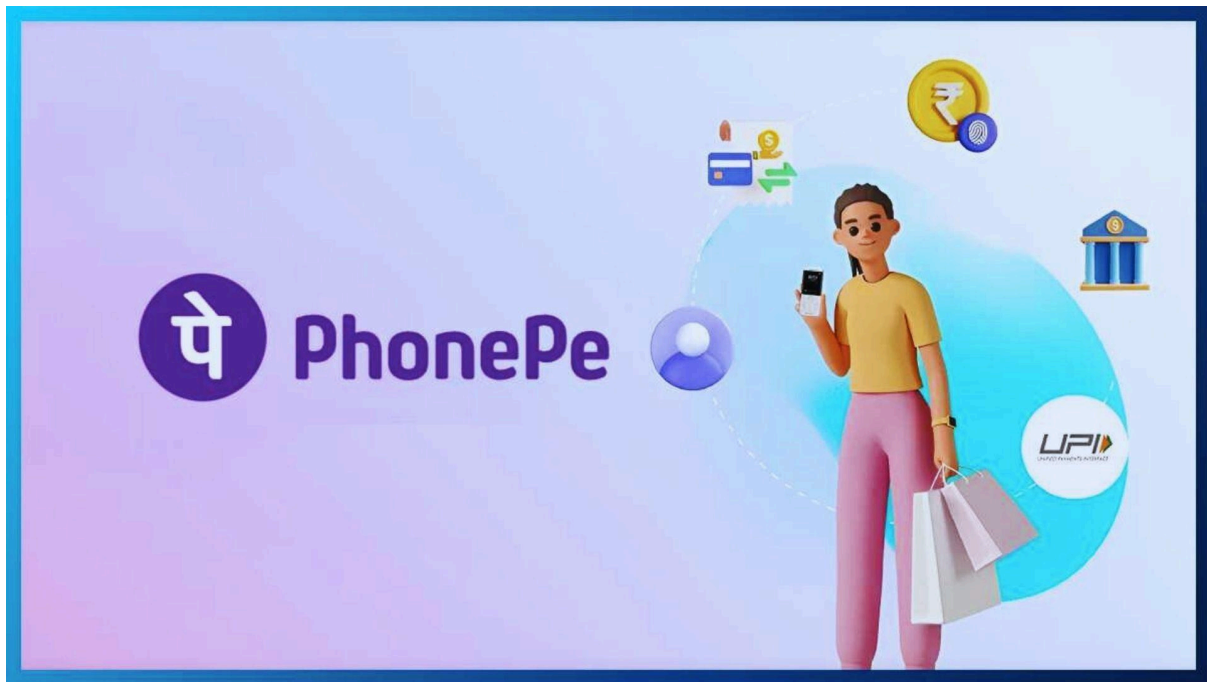
Many online wallets relied on unsustainable cashback structures as a way of keeping users. But this loyalty evaporated once the offers stopped.

PhonePe Solution:

PhonePe was habit and utility-based, not just offers. With high uptime, real use-cases, tie-ups with big billers and banks, and seamless UX, PhonePe builds long-term loyalty on the value, not discounts. It has grown mission-based and organically.

Conclusion:

PhonePe is not just a fintech app but a symbol of Digital India. With its effortless integration of payments, banking, investment, and commerce, PhonePe is solving real money issues of people from different geographies and walks of life in India. From facilitating a farmer in Bihar to make digital payment for seeds to facilitating a startup entrepreneur in Bengaluru to buy health insurance with the click of a button, PhonePe is bringing revolutionary changes in lives. With its user-first approach, secure design, and scalable architecture, PhonePe has emerged as a reliable digital wallet for a billion dreams. It is a beacon of hope for Bharat's move away from cash and towards digital payments and is a made-in-India innovation that is leading the nation towards a financially empowered tomorrow.



Top Features of PhonePe

User Profiles

Every user on PhonePe begins with a secure, personal account tied to their mobile number. This includes optional details like email, full name, and KYC status. It forms the digital identity through which all services, from transactions to investments, are accessed.

UPI Payments

The backbone of PhonePe is India's UPI infrastructure. Users can link their bank accounts using UPI and send/receive money securely and instantly. Each bank account is linked to a user profile and carries UPI IDs, masked account numbers, and IFSC codes.

Recharge & Bill Payments

Users can pay for mobile recharges, electricity, gas, broadband, and other utility bills. Each bill is recorded with its own entity, with a transaction ID linking to the payment made, enabling traceability and transparency.

Merchant Payments (QR Scanning)

PhonePe supports business transactions through QR scanning or UPI ID entry. Merchant profiles store information such as business names, categories, QR code links, and receive payments through associated transactions.

Wallet & Gift Cards

The PhonePe Wallet is a digital balance container used for refunds, cashback, or partial payments. Each wallet is tied to a user, and wallet involvement is recorded in the corresponding transaction.

Offers & Rewards

Scratch cards, cashback, and promotional rewards are granted on eligible transactions. Each reward is linked back to the transaction that triggered it, enabling audit and redemption tracking.

Gold, Mutual Funds & Insurance

PhonePe supports digital investments , from gold to mutual funds and insurance. Each investment is linked to the user and to the transaction through which it was made, enabling detailed financial records.

Transaction History & Notifications

PhonePe maintains a complete record of every transaction, bill payment, and reward. System-generated notifications alert users to transaction outcomes, rewards, and promotional messages.



Schema Description

The PhonePe schema consists of several interconnected entities: Users, BankAccounts, Transactions, Merchants, Bills, Rewards, Investments, Wallets, and Notifications. These entities collectively represent how the app operates, tracking every interaction from money transfers to investment payments.

User Entity

The core identity of each PhonePe user.

- **UserID (Primary Key):** Unique identifier
- **PhoneNumber:** Registered mobile number
- **Full_Name:** Display name
- **Email:** Optional email
- **KYC_Status:** Boolean flag (Yes/No)
- **Registration_Date:** Account creation timestamp

BankAccount Entity

Linked bank accounts for UPI payments.

- **AccountID (Primary Key):** Unique bank account ID
- **UserID (Foreign Key):** Linked user
- **Bank_Name:** Bank name (e.g., SBI, ICICI)
- **Account_Number:** Masked number (e.g., XXXX1234)
- **IFSC_Code:** Bank branch code
- **UPI_ID:** Virtual Payment Address
- **Link_Date:** Date account was added

Merchant Entity

Business profiles for merchant payments.

- **MerchantID (Primary Key):** Unique merchant ID
- **Merchant_Name:** Business/store name
- **UPI_ID:** Merchant's UPI
- **QR_Code_URL:** QR image URL
- **Category:** Type of business (Retail, Services, etc.)
- **Onboard_Date:** When merchant joined

Wallet Entity

Digital wallets for refunds and cashback.

- **WalletID (Primary Key):** One per user
- **UserID (Foreign Key):** Linked user
- **Balance:** Wallet amount in ₹
- **Last_Updated:** Timestamp of last wallet activity

Transaction Entity

Tracks every payment activity, UPI, wallet, investment, and more.

- **TransactionID (Primary Key):** Unique transaction ID
- **UserID (Foreign Key):** Initiator of transaction
- **BankAccountID (Foreign Key):** Bank account used
- **WalletID (Foreign Key):** Nullable; if wallet was used
- **MerchantID (Foreign Key):** Nullable; for merchant payments
- **Receiver_UPI_ID:** Receiver's UPI (user or merchant)
- **Amount:** Transaction amount in ₹
- **Transaction_Type:** Send, Receive, Bill Payment, etc.
- **Status:** Success, Failed, Pending
- **Timestamp:** Date and time of transaction

Bill Entity

Details of every bill or recharge transaction.

- **BillID (Primary Key):** Unique bill ID
- **UserID (Foreign Key):** User who paid
- **Service_Type:** Mobile, DTH, Electricity, etc.
- **Provider_Name:** Jio, Airtel, BESCOM, etc.
- **Account_Number:** Consumer account ID
- **Amount:** Amount paid
- **Status:** Paid or Failed
- **Bill_Date:** Payment date
- **TransactionID (Foreign Key):** Link to associated transaction

Reward Entity

Earned benefits for transactions.

- **RewardID (Primary Key):** Unique reward ID
- **UserID (Foreign Key):** Recipient of reward
- **Reward_Type:** Cashback, Coupon, Gold, etc.
- **Reward_Amount:** Reward value (₹ or %)
- **Valid_Till:** Expiry date
- **Reward_Status:** Available, Claimed, Expired
- **TransactionID (Foreign Key):** Transaction that triggered the reward

Investment Entity

User investments via PhonePe.

- **InvestmentID (Primary Key):** Unique investment ID
- **UserID (Foreign Key):** Investor
- **Investment_Type:** Gold, Mutual Fund, Insurance
- **Amount:** Amount invested
- **Purchase_Date:** Date of investment
- **Maturity_Date:** End date (if applicable)
- **TransactionID (Foreign Key):** Transaction used for investment

Notification Entity

System-generated updates for users.

- **NotificationID (Primary Key):** Unique message ID
- **UserID (Foreign Key):** Recipient
- **Message:** Notification text
- **Type:** Transactional, Promotional, Security, etc.
- **Timestamp:** Date and time sent

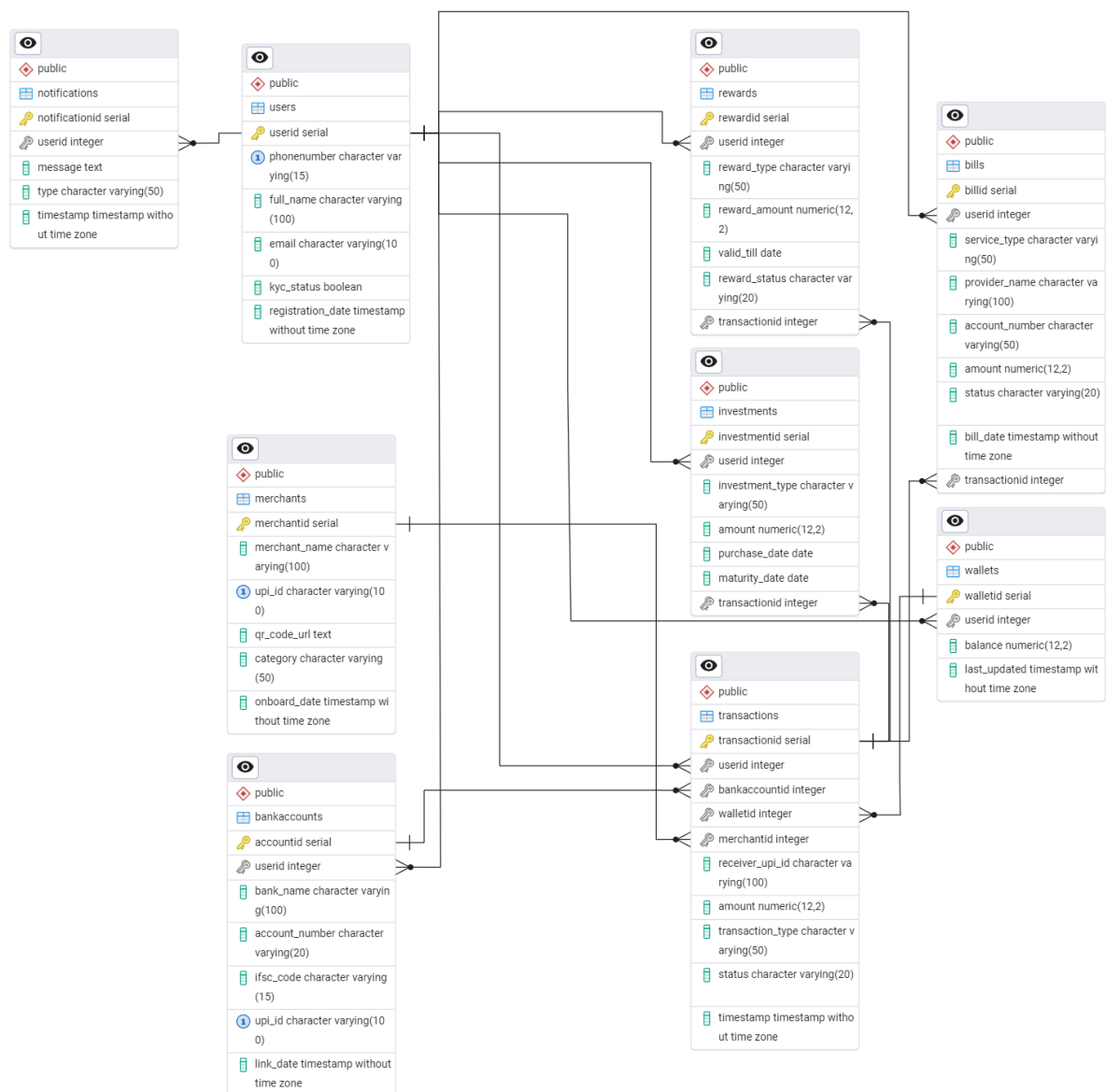
Relationships

- **Users & BankAccounts** :- One user can link multiple accounts
- **Users & Wallets** :- One wallet per user
- **Users & Transactions** :- Users initiate many transactions
- **Transactions & BankAccounts** :- One bank account used per transaction
- **Transactions & Wallets** :- Optional; wallet can be used for payment
- **Transactions & Merchants** :- Payments may be made to merchants
- **Bills & Transactions** :- Every bill is paid through a transaction
- **Rewards & Transactions** :- Rewards are earned via specific transactions
- **Investments & Transactions** :- Investments are funded by transactions
- **Users & Rewards, Bills, Investments, Notifications** :- All linked back to the user



ER Diagram

Let's construct an ER diagram that vividly portrays the relationships and attributes of the entities within the PhonePe schema. This ER diagram will serve as a visual representation, shedding light on the pivotal components of PhonePe's data model. By employing this diagram, we'll gain a clearer grasp of the intricate interactions and connections that define the platform's dynamics.



Conclusion

In conclusion, this detailed product dissection of the PhonePe app showcases the powerful architecture that drives one of India's most trusted digital payment platforms. From seamless UPI transactions and bill payments to merchant services, wallet integration, rewards, and investments, each feature is carefully mapped in the schema with interlinked entities ensuring security, traceability, and scalability. By incorporating robust relationships between users, bank accounts, transactions, and services, this schema lays the foundation for a well-engineered fintech ecosystem. PhonePe's design reflects a vision aligned with India's digital growth, enabling financial inclusion, real-time convenience, and a user-first experience rooted in trust.